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Lecture notes #12 : Solving Dynamic Rational Expectations Models

Franck Portier
(based on Gilles Saint-Paul lecture notes)
franck.portier@TSE-fr.eu

Toulouse School of Economics

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Changes from version 1.0 are in red

Disclaimer

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Introduction

- Economic agents have to make a decision based on their forecast of some future variable.
- In the most general case, the entire probability distribution of that variable will affect their optimal choice.
- The rational expectations hypothesis states that people take into account the *actual* distribution of the relevant variable they want to forecast.
- If expectations are rational, we have to solve simultaneously for the equilibrium distribution and for the rest of the model
- We restrict to linear models in which the distribution of endogenous variables of interest only matters through its first moment, i.e. its mathematical expectation.

1. A simple example

Model

- Agricultural market where producers have to decide how much to produce prior to knowing the equilibrium price

$$
\begin{aligned}
y_t^d &= a - bp_t + \varepsilon_{dt} \\
y_t^s &= c + dp_t^a + \varepsilon_{st}, \\
y_t^s &= y_t^d.
\end{aligned}
$$

- Here, p =price,ε_d=demand shock, ε_s=supply shock,y^s =supply, y^d =demand,p^a = anticipated price.

1. A simple example

Rational expectations

- The Rational Expectations Hypothesis assumes that

$$p_t^a = E(p_t \mid I_t),$$

where $E =$ mathematical expectation operator and $I_t =$ information set of the price-setters when they set y_t^s.

- Typically, $I_t =$ $\{$past values of all the variables, the model, how to solve it$\}$
- If farmers set y_t^s at date $t-1$, :

$$p_t^a = E_{t-1} p_t.$$

1. A simple example

Solving the model

- General principle :
  - take expectations in all the equations and then solve for the expected variables,
  - then substitute this in all the equations where expectations enter,
  - this allows to solve for the actual equilibrium variables.
- Let us try this here.

1. A simple example

Solving the model (continued)

$$
\begin{aligned}
y_t^d &= a - bp_t + \varepsilon_{dt} \\
y_t^s &= c + dp_t^a + \varepsilon_{st}, \\
y_t^s &= y_t^d.
\end{aligned}
$$

- The model can be collapsed to

$$a - bp_t + \varepsilon_{dt} = c + dE_{t-1}p_t + \varepsilon_{st}. \tag{1}$$

- Take expectations :

$$a - bE_{t-1}p_t + E_{t-1}\varepsilon_{dt} = c + dE_{t-1}p_t + E_{t-1}\varepsilon_{st}.$$

- Thus

$$E_{t-1}p_t = \frac{a - c + E_{t-1}(\varepsilon_{dt} - \varepsilon_{st})}{b + d}.$$

1. A simple example

Solving the model (continued)

- substitute this into (1) to solve for p_t :

$$p_t = \frac{1}{b}(\varepsilon_d - \varepsilon_s) - \frac{d}{b(b+d)} E_{t-1}(\varepsilon_d - \varepsilon_s) + \frac{a-c}{b+d}. \qquad (2)$$

- This is the solution.

1. A simple example

The method of undetermined coefficients

- If we know the distribution of those shocks, we can use another method called the method of undetermined coefficients
- This is akin to the notion of Markov perfect equilibrium in game theory.
- Method :
  - Identify the relevant state space
  - Look for a solution as a (linear) function of this state vector
  - Given such a function, I can compute all the expectations I need
  - Subsitute this function and the appropriate formula for expectations in all places where it is needed
  - Identify the coefficients on each state variable
  - This gives us equations in the coefficients that allow us to compute them

1. A simple example

The method of undetermined coefficients (continued)

- Assume $E_{t-1}\varepsilon_{dt} = E_{t-1}\varepsilon_{st} = 0$.
- The relevant state space is $(\varepsilon_d, \varepsilon_s)$.
- Look for a solution of the form

$$p_t = \alpha\varepsilon_{dt} + \beta\varepsilon_{st} + \gamma.$$

1. A simple example

The method of undetermined coefficients (continued)

- Under the assumed solution, we have

$$E_{t-1} p_t = \gamma.$$

- Substituting into (1) we get

$$a - b\left(\alpha\varepsilon_{dt} + \beta\varepsilon_{st} + \gamma\right) + \varepsilon_{dt} = c + d\gamma + \varepsilon_{st}.$$

- This must hold for all realizations of ε_d and ε_s and therefore we must have

$$\begin{aligned} a - b\gamma &= c + d\gamma \\ -\alpha b + 1 &= 0 \\ -\beta b &= 1. \end{aligned}$$

1. A simple example

The method of undetermined coefficients (continued)

- Therefore

$$\begin{aligned}\gamma &= \frac{a-c}{b+d} \\ \alpha &= 1/b \\ \beta &= -1/b.\end{aligned}$$

- Hence the solution

$$p_t = \frac{1}{b}(\varepsilon_d - \varepsilon_s) + \frac{a-c}{b+d},$$

- which is equivalent to (2) if $E_{t-1}\varepsilon_{dt} = E_{t-1}\varepsilon_{st} = 0$.

2. A dynamic model

Model

- Suppose now the supply curve is

$$y_t^s = c + dE_t p_{t+1} + \varepsilon_{st}$$

- Eliminating y between supply and demand :

$$p_t = \frac{a-c}{b} + \frac{-d}{b} E_t p_{t+1} + \frac{\varepsilon_{dt} - \varepsilon_{st}}{b}$$

- and assuming for simplicity that the constant terms are zero, we end up with

$$p_t = \rho E_t p_{t+1} + \eta_t, \tag{3}$$

where η_t is a shock.

2. A dynamic model

Solution by forward integration

- (3) implies $p_{t+i} = \rho E_t p_{t+i+1} + \eta_t \quad \forall i$
- Taking expectations

$$E_t p_{t+i} = \rho E_t p_{t+i+1} + E_t \eta_{t+i}$$

- Substitute forward (*forward integration*) :

$$p_t = \rho^k E_t p_{t+k} + \sum_{j=0}^{k-1} \rho^j E_t \eta_{t+j} \tag{4}$$

- Assume stationary shocks and impose $E_t p_{t+k}$ is bounded $\forall k$.
- (stochastic equivalent of selecting the non-explosive saddle-path as the unique solution.)
- Then if $\mid \rho \mid < 1$, then $\lim_{k\to\infty} \rho^k E_t p_{t+k} = 0$, hence

$$p_t = \sum_{j=0}^{\infty} \rho^j E_t \eta_{t+j}.$$

2. A dynamic model

Solution by forward integration (continued)

$$p_t = \sum_{j=0}^{\infty} \rho^j E_t \eta_{t+j}.$$

- This is the unique stationary solution to the rational expectations equation (3).
- Suppose $\mid \rho \mid > 1$, then the solution cannot be unique.
- For any solution p_t, $p_t + x_t$ is also a solution under some conditions on x
- x_t is a random variable (sunspot) satisfying

$$x_{t+1} = \frac{1}{\rho} x_t + u_{t+1}, \tag{5}$$

  with u any shock such that $E_t u_{t+1} = 0$.
- Furthermore, we can construct at least one non explosive solution by picking $p_{t+1} = \frac{1}{\rho}(p_t - \eta_t)$.
- Hence in this case we have a continuum of equilibria.

2. A dynamic model

Undetermined coefficients

- We can apply the method of undetermined coefficients to this problem if we know the distribution of η.
- Assume η follows an AR1 :

$$\eta_t = \alpha\eta_{t-1} + \omega_t,$$

  with ω_t iid and with zero mean.
- The only relevant state variables are η_{t-1} and ω_t so we look for a solution of the form

$$p_t = \lambda\eta_{t-1} + \beta\omega_t.$$

- Then

$$\begin{aligned} E_t p_{t+1} &= E_t(\lambda\eta_t + \beta\omega_{t+1}) \\ &= \lambda\eta_t \\ &= \lambda(\alpha\eta_{t-1} + \omega_t). \end{aligned}$$

2. A dynamic model

Undetermined coefficients (continued)

- We can then rewrite (3) as

$$\lambda\eta_{t-1} + \beta\omega_t = \rho\lambda(\alpha\eta_{t-1} + \omega_t) + \alpha\eta_{t-1} + \omega_t.$$

- Identifying, we get

$$\begin{aligned} \lambda &= \rho\lambda\alpha + \alpha \\ \beta &= \rho\lambda + 1 \end{aligned}$$

- Hence

$$\begin{aligned} \lambda &= \frac{\alpha}{1-\rho\alpha}; \\ \beta &= \frac{1}{1-\rho\alpha}. \end{aligned}$$

2. A dynamic model

Undetermined coefficients (continued)

- We can check that if $\mid \rho \mid < 1$, this is indeed correct :

$$\begin{aligned}
\sum_{j=0}^{\infty} \rho^j E_t \eta_{t+j}. &= \sum_{j=0}^{\infty} \rho^j \alpha^j \eta_t \\
&= \frac{1}{1-\rho\alpha}(\alpha\eta_{t-1} + \omega_t) \\
&= \lambda\eta_{t-1} + \beta\omega_t
\end{aligned}$$

- If $\mid \rho \mid > 1$, this is still a solution, and it is the unique "linear Markov" solution, i.e. the only one which is a linear combination of the relevant state space.
- But many other solutions can be constructed by adding a sunspot x satisfying (5).

- General linear model (Blanchard-Kahn)

$$\begin{pmatrix} K_t \\ X_t \end{pmatrix} = A \begin{pmatrix} K_{t-1} \\ E_t X_{t+1} \end{pmatrix} + U_t, \qquad (6)$$

- K_t is an $m \times 1$ vector of state (=predetermined) variables, meaning that K_{t-1} is fixed at the beginning of period t,
- X_t is an $n \times 1$ vector of non-predetermined variables, i.e. expected future values of those variables, rather than lagged ones, affect current outcomes.
- A is an $(m+n) \times (m+n)$ matrix
- U an $(m+n) \times 1$ vector of shocks.

3. A general model

Model transformation

- Let
$$A = \begin{pmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{pmatrix} \quad , \qquad U_t = \begin{pmatrix} U_{0t} \\ U_{1t} \end{pmatrix},$$
- then
$$\begin{aligned} K_t &= A_{00} K_{t-1} + A_{01} E_t X_{t+1} + U_{0t} \\ X_t &= A_{10} K_{t-1} + A_{11} E_t X_{t+1} + U_{1t}. \end{aligned}$$
- We can solve for $E_t X_{t+1}$ and write the whole system with backward-looking dynamics :

$$\begin{aligned} \begin{pmatrix} K_t \\ E_t X_{t+1} \end{pmatrix} &= \begin{pmatrix} A_{00} - A_{01} A_{11}^{-1} A_{10} & A_{01} A_{11}^{-1} \\ -A_{11}^{-1} A_{10} & A_{11}^{-1} \end{pmatrix} \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix} \\ &+ \begin{pmatrix} U_{0t} - A_{01} A_{11}^{-1} U_{1t} \\ -A_{11}^{-1} U_{1t} \end{pmatrix} \\ &= M \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix} + V_t. \end{aligned} \tag{7}$$

3. A general model

The diagonal case

$$\begin{pmatrix} K_t \\ E_t X_{t+1} \end{pmatrix} = M \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix} + V_t. \qquad (7)$$

- Let us start with the simple case where M is diagonal, say $M = \begin{pmatrix} D_0 & 0 \\ 0 & D_1 \end{pmatrix}$, where $D_0 = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & \dots \\ \dots & \dots & \dots & 0 \\ 0 & \dots & 0 & d_m \end{pmatrix}$ and $D_1 = \begin{pmatrix} d'_1 & 0 & \dots & 0 \\ 0 & d'_2 & \dots & \dots \\ \dots & \dots & \dots & 0 \\ 0 & \dots & 0 & d'_n \end{pmatrix}$ and $V_t = \begin{pmatrix} V_{0t} \\ V_{1t} \end{pmatrix}$

3. A general model

The diagonal case (continued)

- Then, iterating (7),

$$K_{t+i} = D_0^{i+1} K_{t-1} + \sum_{j=0}^{i} D_0^{i-j} V_{0t+j}.$$

- For K_{t+i} to remain bounded, we need

$$\mid d_i \mid < 1, i = 1, ..., m. \tag{8}$$

- Similarly

$$\begin{aligned} E_t X_{t+i} &= D_1^i X_t + \sum_{j=0}^{i-1} D_1^{i-1-j} E_t V_{1t+j} \\ &= D_1^i (X_t + \sum_{j=0}^{i-1} D_1^{-1-j} E_t V_{1t+j}). \end{aligned}$$

3. A general model

The diagonal case (continued)

- Consider first a k such that $\mid d_k' \mid < 1$. Let x_{kt} (resp. v_{kt}) be the k-th term in X_t (resp. V_{1t}).
- Then

$$
\begin{aligned}
E_t x_{t+i} &= (d_k')^i x_t + \sum_{j=0}^{i-1} (d_k')^{i-1-j} E_t v_{k,t+j} \\
&\iff (d_k'^{-1})^i E_t x_{t+i} - \sum_{j=0}^{i-1} (d_k'^{-1})^{1+j} E_t v_{k,t+j} = x_t.
\end{aligned}
$$

- This is similar to (4) and since $\mid d_k'^{-1} \mid > 1$ there is no unique initial value of x_t yielding a non-explosive path for subsequent expectations.

3. A general model

The diagonal case (continued)

- Second, assume conversely that

$$\mid d'_k \mid > 1 \text{ for } k = 1, ..., n. \tag{9}$$

- Then all the terms in D_1^i go to infinity as $i \to \infty$.
- Thus it must be that $\lim_{i \to +\infty} X_t + \sum_{j=0}^{i-1} D_1^{-1-j} E_t V_{1t+j} = 0$. Therefore,

$$X_t = -\sum_{j=0}^{\infty} \left(D_1^{-1}\right)^{1+j} E_t V_{1t+j}.$$

3. A general model

The diagonal case (continued)

$$\mid d_i \mid < 1, i = 1, ..., m. \tag{8}$$

$$\mid d'_k \mid > 1 \text{ for } k = 1, ..., n. \tag{9}$$

- To summarize
  - To have a unique non-explosive solution, (8) and (9) must hold
  - If (8) is violated, all solutions are explosive
  - If (9) is violated, there is a continuum of non-explosive solutions.

3. A general model

The non diagonal case

- Assume now that the matrix M is not diagonal.
- The idea is to transform the model into a diagonal one.
- Generically, M is diagonalizable in the complex plane $\mathbb{C}$.
- Thus there exists P invertible such that

$$M = P^{-1}DP,$$

with D diagonal :

$$D = \begin{pmatrix} d_0 & 0 & ... & ... & ... & ... & 0 \\ 0 & ... & ... & ... & ... & ... & ... \\ ... & ... & d_m & 0 & ... & ... & ... \\ ... & ... & 0 & d_{m+1} & 0 & ... & ... \\ ... & ... & ... & 0 & ... & ... & ... \\ ... & ... & ... & ... & ... & ... & 0 \\ 0 & ... & ... & ... & ... & 0 & d_{m+n} \end{pmatrix} = \begin{pmatrix} D_0 & 0 \\ 0 & D_1 \end{pmatrix}$$

3. A general model

The non diagonal case (continued)

- The diagonal coefficients are the eigenvalues of M.
- Eigenvalues are ranked by ascending module :
  $\mid d_0 \mid \leq \mid d_1 \mid \leq ... \leq \mid d_m \mid \leq \mid d'_1 \mid \leq \mid d'_2 \mid \leq ... \leq \mid d'_n \mid$.
- (7) rewrite

$$P \begin{pmatrix} K_t \\ E_t X_{t+1} \end{pmatrix} = D \left(P \begin{pmatrix} K_t \\ X_t \end{pmatrix} \right) + PV_t .$$

- Let $Z_t = P \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix}$;
- We are back to the diagonal case with this change of variable.

4. How general is the general model ?

More lags for K

- Suppose that we have terms in K_{t-k}, $k > 1$.
- The solution is to add the variables $(K_{t-1}, ..., K_{t-k+1})$ to the state vector.
- Suppose we have this equation $K_t = aK_{t-1} + bK_{t-k}$
- Let us assume to save on notation that K is a scalar.
- Then the vector $\begin{pmatrix} K_t \\ K_{t-1} \\ ... \\ K_{t-k+1} \end{pmatrix}$ satisfies

$$\begin{pmatrix} K_t \\ K_{t-1} \\ ... \\ K_{t-k+1} \end{pmatrix} = \begin{pmatrix} a & 0 & ... & b \\ 1 & 0 & ... & ... \\ 0 & ... & ... & ... \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} K_{t-1} \\ K_{t-2} \\ ... \\ K_{t-k} \end{pmatrix},$$

which only involves the first lag of this vector.

4. How general is the general model?

More lags for X

- Suppose we have terms in $E_t X_{t+k}$, $k > 1$.
- Let's add $E_t X_{t+1}, \ldots, E_t X_{t+k-1}$ to the jump variables
- Suppose that we have

$$X_t = aE_t X_{t+1} + bE_t X_{t+k}.$$

- Then the vector $\Psi_t = (X_t, E_t X_{t+1}, \ldots, E_t X_{t+k-1})'$ satisfies

$$\begin{aligned}
\Psi_t &= \begin{pmatrix} X_t \\ E_t X_{t+1} \\ \ldots \\ E_t X_{t+k-1} \end{pmatrix} = \begin{pmatrix} a & 0 & \ldots & b \\ 1 & 0 & \ldots & \ldots \\ 0 & \ldots & \ldots & \ldots \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} E_t X_{t+1} \\ E_t X_{t+2} \\ \ldots \\ E_t X_{t+k} \end{pmatrix} \\
&= \begin{pmatrix} a & 0 & \ldots & b \\ 1 & 0 & \ldots & \ldots \\ 0 & \ldots & \ldots & \ldots \\ 0 & 0 & 1 & 0 \end{pmatrix} E_t \Psi_{t+1}.
\end{aligned}$$